LINEAR ALGEBRA • UNIT 8.01

C Orthogonality, Gram-Schmidt, QR & Least Squares

ANSWER KEY & SOLUTIONS

25 Structured Exam-Style Questions

Total Questions: 25 • Verified Step-by-Step Solutions

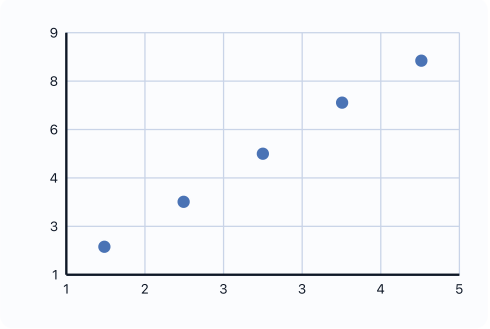
QUICK-REFERENCE ANSWER SUMMARY (25 ITEMS)

Q1 B	Q2 B	Q3 C	Q4 A	Q5 B
Q6 D	Q7 D	Q8 C	Q9 C	Q10 D
Q11 C	Q12 A	Q13 B	Q14 D	Q15 B
Q16 D	Q17 A	Q18 B	Q19 C	Q20 A
Q21 D	Q22 C	Q23 C	Q24 A	Q25 C

Orthogonality, Gram-Schmidt, QR & Least Squares Practice Exercises

1 MCQ

For which k are $\langle 0, -2, -2\rangle$ and $\langle -1, 2, k\rangle$ orthogonal?



A 2

B -2 ✓

C -1

D 3

WORKED SOLUTION & VERIFICATION

Set the dot product to zero: $-2k - 4 = 0$, so $k = -2$.

2 MCQ

Which set is NOT orthonormal?

A $\{\langle \frac{5}{13}, \frac{12}{13}\rangle, \langle \frac{12}{13}, -\frac{5}{13}\rangle\}$

B $\{\langle \frac{1}{2}, \frac{1}{2}\rangle, \langle \frac{1}{2}, -\frac{1}{2}\rangle\}$ ✓

C $\{\langle \frac{1}{3}, \frac{2}{3}, \frac{2}{3}\rangle, \langle \frac{2}{3}, \frac{1}{3}, -\frac{2}{3}\rangle\}$

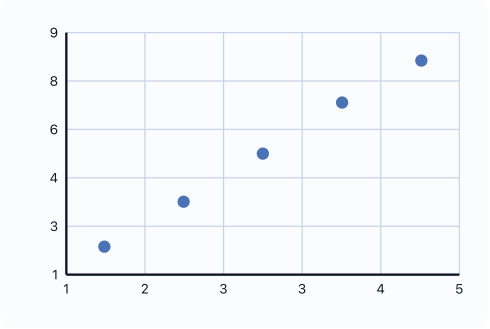
D $\{\langle \frac{\sqrt{2}}{2}, \frac{\sqrt{2}}{2}\rangle, \langle \frac{\sqrt{2}}{2}, -\frac{\sqrt{2}}{2}\rangle\}$

WORKED SOLUTION & VERIFICATION

$\{\langle \frac{1}{2}, \frac{1}{2}\rangle, \langle \frac{1}{2}, -\frac{1}{2}\rangle\}$ fails: either the dot product is nonzero or a vector does not have length 1. The other sets pass both tests.

3 MCQ

Which matrix is orthogonal (satisfies $Q^TQ = I$)?



A $\begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$

B $\begin{bmatrix} \frac{1}{2} & \frac{1}{2} \\ \frac{1}{2} & -\frac{1}{2} \end{bmatrix}$

C $\begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$ ✓

D $\begin{bmatrix} 1 & -1 \\ 1 & 1 \end{bmatrix}$

WORKED SOLUTION & VERIFICATION

The columns of $\begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$ are orthonormal, so $Q^TQ = I$. Each other matrix has columns that are not orthogonal or not unit length.

4 MCQ

Normalize $\mathbf{v} = \langle 6, 8 \rangle$.

A $\langle \frac{3}{5}, \frac{4}{5} \rangle$ ✓

B $\langle \frac{3}{50}, \frac{2}{25} \rangle$

C $\langle \frac{3}{7}, \frac{4}{7} \rangle$

D $\langle -\frac{3}{5}, -\frac{4}{5} \rangle$

WORKED SOLUTION & VERIFICATION

$\|\mathbf{v}\| = 10$, so $\mathbf{v}/\|\mathbf{v}\| = \langle \frac{3}{5}, \frac{4}{5} \rangle$.

5 MCQ

Which set is orthonormal?

A $\{\langle 1, 2, 2 \rangle, \langle 2, 1, -2 \rangle\}$

B $\{\langle \frac{1}{3}, \frac{2}{3}, \frac{2}{3} \rangle, \langle \frac{2}{3}, \frac{1}{3}, -\frac{2}{3} \rangle\}$ ✓

C $\{\langle \frac{1}{2}, \frac{1}{2} \rangle, \langle \frac{1}{2}, -\frac{1}{2} \rangle\}$

D $\{\langle \frac{3}{5}, \frac{4}{5} \rangle, \langle \frac{4}{5}, \frac{3}{5} \rangle\}$

WORKED SOLUTION & VERIFICATION

$\{\langle \frac{1}{3}, \frac{2}{3}, \frac{2}{3} \rangle, \langle \frac{2}{3}, \frac{1}{3}, -\frac{2}{3} \rangle\}$: the vectors are orthogonal and each has length 1. Each other set fails one of these tests.

6 MCQ

For which k are $\langle 3, 3, 2\rangle$ and $\langle 2, k, 1\rangle$ orthogonal?

A $\frac{8}{3}$

B $-\frac{5}{3}$

C $\frac{11}{3}$

D $-\frac{8}{3}$ ✓

WORKED SOLUTION & VERIFICATION

Set the dot product to zero: $3k + 8 = 0$, so $k = -\frac{8}{3}$.

7 MCQ

Normalize $\mathbf{v} = \langle -2, 1, -2\rangle$.

A $\langle -\frac{2}{9}, \frac{1}{9}, -\frac{2}{9}\rangle$

B $\langle -\frac{2}{5}, \frac{1}{5}, -\frac{2}{5}\rangle$

C $\langle \frac{2}{3}, -\frac{1}{3}, \frac{2}{3}\rangle$

D $\langle -\frac{2}{3}, \frac{1}{3}, -\frac{2}{3}\rangle$ ✓

WORKED SOLUTION & VERIFICATION

$\|\mathbf{v}\| = 3$, so $\mathbf{v}/\|\mathbf{v}\| = \langle -\frac{2}{3}, \frac{1}{3}, -\frac{2}{3}\rangle$.

8 MCQ

For which k are $\langle 0, -3, 1\rangle$ and $\langle 1, k, -1\rangle$ orthogonal?

A $\frac{1}{3}$

B $\frac{2}{3}$

C $-\frac{1}{3}$ ✓

D $\frac{4}{3}$

WORKED SOLUTION & VERIFICATION

Set the dot product to zero: $-3k - 1 = 0$, so $k = -\frac{1}{3}$.

9 MCQ

Consider two vectors $\mathbf{v}_1 = \begin{pmatrix} 3 \\ 4 \end{pmatrix}$ and $\mathbf{v}_2 = \begin{pmatrix} -4 \\ 3 \end{pmatrix}$. What is the dot product $\mathbf{v}_1 \cdot \mathbf{v}_2$?

A 1

B -7

C 0 ✓

D 25

WORKED SOLUTION & VERIFICATION

The dot product of two vectors $\mathbf{v}_1 = \begin{pmatrix} a \\ b \end{pmatrix}$ and $\mathbf{v}_2 = \begin{pmatrix} c \\ d \end{pmatrix}$ is given by $\mathbf{v}_1 \cdot \mathbf{v}_2 = ac + bd$.

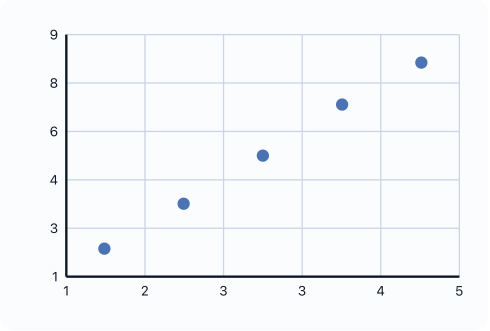
In this case, $\mathbf{v}_1 = \begin{pmatrix} 3 \\ 4 \end{pmatrix}$ and $\mathbf{v}_2 = \begin{pmatrix} -4 \\ 3 \end{pmatrix}$.

Therefore, $\mathbf{v}_1 \cdot \mathbf{v}_2 = (3)(-4) + (4)(3) = -12 + 12 = 0$.

A dot product of 0 indicates that the vectors are orthogonal.

10 MCQ

Consider the matrix $A = \begin{pmatrix} 1 & 2 \\ -2 & k \end{pmatrix}$. If the column vectors of A are orthogonal, find the value of k.



A 4

B -2

C -1

D 2 ✓

WORKED SOLUTION & VERIFICATION

Let the column vectors of matrix A be $\mathbf{v}_1 = \begin{pmatrix} 1 \\ -2 \end{pmatrix}$ and $\mathbf{v}_2 = \begin{pmatrix} 2 \\ k \end{pmatrix}$. \nFor the column vectors to be orthogonal, their dot product must be zero. \n$\mathbf{v}_1 \cdot \mathbf{v}_2 = 0$ \n$(1)(2) + (-2)(k) = 0$ \n$2 - 2k = 0$ \n$2k = 2$ \n$k = 1$.

11 MCQ

Consider the matrix $A = \begin{pmatrix} 1 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 1 \end{pmatrix}$. Let $\mathbf{b} = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}$. We want to find the vector $\hat{\mathbf{x}}$ that minimizes $\|A\mathbf{x} - \mathbf{b}\|$. This is equivalent to finding the projection of $\mathbf{b}$ onto the column space of A, denoted by $\text{Col}(A)$. The normal equations for this least squares problem are $A^T A\mathbf{x} = A^T\mathbf{b}$. The unique solution $\hat{\mathbf{x}}$ to the normal equations yields the best approximate solution. Calculate the value of $\hat{\mathbf{x}}_1$, the first component of $\hat{\mathbf{x}}$.

A 1/3

B 2/3

C 1/2 ✓

D 3/4

WORKED SOLUTION & VERIFICATION

The normal equations are given by $A^T A\mathbf{x} = A^T\mathbf{b}$. First, we compute $A^T A$ and $A^T\mathbf{b}$.

$$A^T = \begin{pmatrix} 1 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 1 \end{pmatrix}$$

$$A^T A = \begin{pmatrix} 1 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 1 \end{pmatrix} \begin{pmatrix} 1 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 1 \end{pmatrix} = \begin{pmatrix} 2 & 1 & 1 \\ 1 & 2 & 1 \\ 1 & 1 & 2 \end{pmatrix}$$

$$A^T\mathbf{b} = \begin{pmatrix} 1 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 1 \end{pmatrix} \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} = \begin{pmatrix} 2 \\ 2 \\ 2 \end{pmatrix}$$

So, we need to solve the system: $\begin{pmatrix} 2 & 1 & 1 \\ 1 & 2 & 1 \\ 1 & 1 & 2 \end{pmatrix} \hat{\mathbf{x}} = \begin{pmatrix} 2 \\ 2 \\ 2 \end{pmatrix}$

From the symmetry of the matrix $A^T A$ and the vector $A^T\mathbf{b}$, it is clear that $\hat{\mathbf{x}} = \begin{pmatrix} c \\ c \\ c \end{pmatrix}$ for some constant c.

Substituting this into the first row of the system: $2c + 1c + 1c = 2$

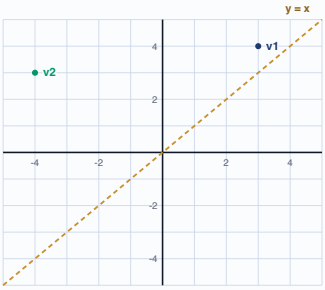
$4c = 2$

$c = \frac{1}{2}$

Thus, $\hat{\mathbf{x}} = \begin{pmatrix} 1/2 \\ 1/2 \\ 1/2 \end{pmatrix}$. The first component of $\hat{\mathbf{x}}$ is $\hat{\mathbf{x}}_1 = 1/2$.

12 MCQ

Consider the vectors $\mathbf{v}_1 = \begin{pmatrix} 3 \\ 4 \end{pmatrix}$ and $\mathbf{v}_2 = \begin{pmatrix} -4 \\ 3 \end{pmatrix}$. What is the dot product $\mathbf{v}_1 \cdot \mathbf{v}_2$?



- **A** 0 ✓
- **B** 1
- **C** 7
- **D** -7

WORKED SOLUTION & VERIFICATION

To find the dot product of two vectors $\mathbf{v}_1 = \begin{pmatrix} a \\ b \end{pmatrix}$ and $\mathbf{v}_2 = \begin{pmatrix} c \\ d \end{pmatrix}$, we compute $\mathbf{v}_1 \cdot \mathbf{v}_2 = ac + bd$. In this case, $\mathbf{v}_1 = \begin{pmatrix} 3 \\ 4 \end{pmatrix}$ and $\mathbf{v}_2 = \begin{pmatrix} -4 \\ 3 \end{pmatrix}$. Therefore, the dot product is $(3)(-4) + (4)(3) = -12 + 12 = 0$. Since the dot product is 0, the vectors are orthogonal.

13 MCQ

A 3×3 symmetric matrix T_a has eigenvalues $\lambda_1 = 9$, $\lambda_2 = 24$, and $\lambda_3 = 20$. Determine the trace $\text{tr}(T_a)$.

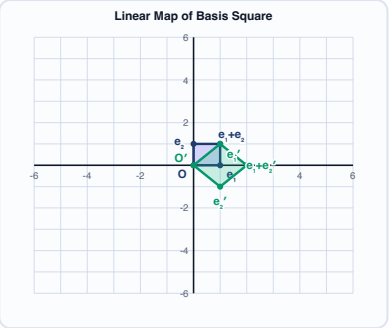
- **A** 29
- **B** 53 ✓
- **C** 56
- **D** 62

WORKED SOLUTION & VERIFICATION

The trace equals $\lambda_1 + \lambda_2 + \lambda_3 = 53$.

14 MCQ

Consider the matrix $A = \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$. Let $\mathbf{v} = \begin{pmatrix} 3 \\ 1 \end{pmatrix}$. Suppose we want to find the vector $\mathbf{x}$ of the form $c_1\mathbf{a}_1 + c_2\mathbf{a}_2$ where $\{\mathbf{a}_1, \mathbf{a}_2\}$ is an orthogonal basis derived from the columns of A, such that $\|\mathbf{v} - \mathbf{x}\|$ is minimized. Find the value of $\|\mathbf{x}\|$.



A $2\sqrt{10}$

B $\sqrt{20}$

C $\sqrt{5}$

D $\sqrt{10}$ ✓

WORKED SOLUTION & VERIFICATION

The columns of matrix A are $\mathbf{c}_1 = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$ and $\mathbf{c}_2 = \begin{pmatrix} 1 \\ -1 \end{pmatrix}$. This set of vectors is already orthogonal since $\mathbf{c}_1 \cdot \mathbf{c}_2 = (1)(1) + (1)(-1) = 0$. Thus, $\{\mathbf{a}_1, \mathbf{a}_2\} = \{\mathbf{c}_1, \mathbf{c}_2\}$ is an orthogonal basis. The problem asks to find the projection of $\mathbf{v}$ onto the subspace spanned by $\mathbf{a}_1$ and $\mathbf{a}_2$, which is the entire R^2 in this case, as $\{\mathbf{a}_1, \mathbf{a}_2\}$ forms a basis for R^2. Therefore, the vector $\mathbf{x}$ that minimizes $\|\mathbf{v} - \mathbf{x}\|$ is simply $\mathbf{v}$ itself, as $\mathbf{v}$ is already in the span of $\mathbf{a}_1$ and $\mathbf{a}_2$. So, $\mathbf{x} = \mathbf{v} = \begin{pmatrix} 3 \\ 1 \end{pmatrix}$. The norm of $\mathbf{x}$ is $\|\mathbf{x}\| = \|\mathbf{v}\| = \sqrt{3^2 + 1^2} = \sqrt{9 + 1} = \sqrt{10}$.

15 MCQ

Consider the vector space R^2 with the standard dot product. Let $\mathbf{u} = (2, 3)$ and $\mathbf{v} = (-3, 2)$. Which of the following statements about $\mathbf{u}$ and $\mathbf{v}$ is true?

A $\mathbf{u}$ and $\mathbf{v}$ are linearly dependent.

B **$\mathbf{u}$ and $\mathbf{v}$ are orthogonal.** ✓

C The angle between $\mathbf{u}$ and $\mathbf{v}$ is acute.

D The angle between $\mathbf{u}$ and $\mathbf{v}$ is obtuse.

WORKED SOLUTION & VERIFICATION

To determine if two vectors are orthogonal, we compute their dot product. If the dot product is zero, the vectors are orthogonal.

For the given vectors $\mathbf{u} = (2, 3)$ and $\mathbf{v} = (-3, 2)$, the dot product is:

$\mathbf{u} \cdot \mathbf{v} = (2)(-3) + (3)(2) = -6 + 6 = 0$.

Since the dot product $\mathbf{u} \cdot \mathbf{v} = 0$, the vectors $\mathbf{u}$ and $\mathbf{v}$ are orthogonal.

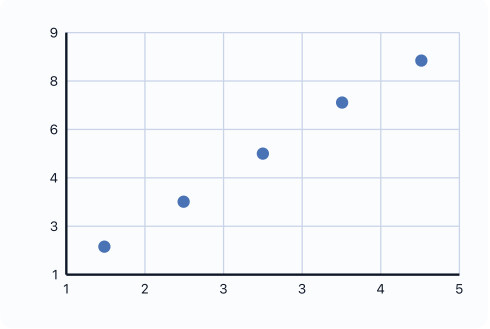
Therefore, the statement "$\mathbf{u}$ and $\mathbf{v}$ are orthogonal" is true.

If the dot product were positive, the angle would be acute. If the dot product were negative, the angle would be obtuse. Since the dot product is zero, the angle is exactly 90 degrees, which is neither acute nor obtuse in the typical classification of angles within a triangle, but implies orthogonality.

Two non-zero vectors are linearly dependent if and only if one is a scalar multiple of the other. $\mathbf{u} = (2, 3)$ and $\mathbf{v} = (-3, 2)$. There is no scalar c such that $(2, 3) = c(-3, 2)$ because $2 = -3c$ implies $c = -2/3$, while $3 = 2c$ implies $c = 3/2$. Since these values of c are not equal, the vectors are not linearly dependent.

16 MCQ

Apply Gram–Schmidt to $\mathbf{x}_1 = \langle 1, -1 \rangle$, $\mathbf{x}_2 = \langle 2, 3 \rangle$. Which vector is parallel to the resulting $\mathbf{v}_2$?



A $\langle 2, 3 \rangle$

B $\langle \frac{3}{2}, \frac{7}{2} \rangle$

C $\langle 1, -1 \rangle$

D $\langle 1, 1 \rangle$ ✓

WORKED SOLUTION & VERIFICATION

$\mathbf{v}_2 = \mathbf{x}_2 - \dfrac{\mathbf{x}_2 \cdot \mathbf{x}_1}{\mathbf{x}_1 \cdot \mathbf{x}_1}\mathbf{x}_1 = \langle \frac{5}{2}, \frac{5}{2} \rangle$, a multiple of $\langle 1, 1 \rangle$.

17 MCQ

Apply Gram–Schmidt to $\mathbf{x}_1 = \langle 3, 2 \rangle$, $\mathbf{x}_2 = \langle 0, 2 \rangle$. Which vector is parallel to the resulting $\mathbf{v}_2$?

A $\langle -2, 3 \rangle$ ✓

B $\langle 0, 2 \rangle$

C $\langle \frac{12}{13}, \frac{34}{13} \rangle$

D $\langle 3, 2 \rangle$

WORKED SOLUTION & VERIFICATION

$\mathbf{v}_2 = \mathbf{x}_2 - \dfrac{\mathbf{x}_2 \cdot \mathbf{x}_1}{\mathbf{x}_1 \cdot \mathbf{x}_1}\mathbf{x}_1 = \langle -\frac{12}{13}, \frac{18}{13} \rangle$, a multiple of $\langle -2, 3 \rangle$.

18 MCQ

Apply Gram–Schmidt to $\mathbf{x}_1 = \langle 1, 1 \rangle$, $\mathbf{x}_2 = \langle 2, -2 \rangle$. Which vector is parallel to the resulting $\mathbf{v}_2$?

A $\langle -1, 1 \rangle$

B $\langle 1, -1 \rangle$ ✓

C $\langle 2, -2 \rangle$

D $\langle 1, 1 \rangle$

WORKED SOLUTION & VERIFICATION

$\mathbf{v}_2 = \mathbf{x}_2 - \dfrac{\mathbf{x}_2 \cdot \mathbf{x}_1}{\mathbf{x}_1 \cdot \mathbf{x}_1}\mathbf{x}_1 = \langle 2, -2 \rangle$, a multiple of $\langle 1, -1 \rangle$.

19 MCQ

Apply Gram–Schmidt to $\mathbf{x}_1 = \langle -1, 3, 1 \rangle$, $\mathbf{x}_2 = \langle 3, -3, -2 \rangle$. Which vector is parallel to the resulting $\mathbf{v}_2$?

A $\langle -1, 3, 1 \rangle$

B $\langle 3, -3, -2 \rangle$

C $\langle 19, 9, -8 \rangle$ ✓

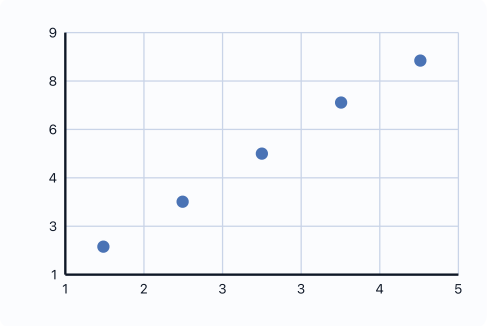
D $\langle \frac{47}{11}, -\frac{75}{11}, -\frac{36}{11} \rangle$

WORKED SOLUTION & VERIFICATION

$\mathbf{v}_2 = \mathbf{x}_2 - \dfrac{\mathbf{x}_2 \cdot \mathbf{x}_1}{\mathbf{x}_1 \cdot \mathbf{x}_1}\mathbf{x}_1 = \langle \frac{19}{11}, \frac{9}{11}, -\frac{8}{11} \rangle$, a multiple of $\langle 19, 9, -8 \rangle$.

20 MCQ

Apply Gram–Schmidt to $\mathbf{x}_1 = \langle -1, 3\rangle$, $\mathbf{x}_2 = \langle -3, 2\rangle$. Which vector is parallel to the resulting $\mathbf{v}_2$?



- **A** $\langle -3, -1\rangle$ ✓
- **B** $\langle -3, 2\rangle$
- **C** $\langle -\frac{39}{10}, \frac{47}{10}\rangle$
- **D** $\langle -1, 3\rangle$

WORKED SOLUTION & VERIFICATION

$\mathbf{v}_2 = \mathbf{x}_2 - \dfrac{\mathbf{x}_2 \cdot \mathbf{x}_1}{\mathbf{x}_1 \cdot \mathbf{x}_1}\mathbf{x}_1 = \langle -\frac{21}{10}, -\frac{7}{10}\rangle$, a multiple of $\langle -3, -1\rangle$.

21 MCQ

Apply Gram–Schmidt to $\mathbf{x}_1 = \langle 2, -1, 0\rangle$, $\mathbf{x}_2 = \langle 0, 3, -1\rangle$. Which vector is parallel to the resulting $\mathbf{v}_2$?

- **A** $\langle 0, 3, -1\rangle$
- **B** $\langle -\frac{6}{5}, \frac{18}{5}, -1\rangle$
- **C** $\langle 2, -1, 0\rangle$
- **D** $\langle 6, 12, -5\rangle$ ✓

WORKED SOLUTION & VERIFICATION

$\mathbf{v}_2 = \mathbf{x}_2 - \dfrac{\mathbf{x}_2 \cdot \mathbf{x}_1}{\mathbf{x}_1 \cdot \mathbf{x}_1}\mathbf{x}_1 = \langle \frac{6}{5}, \frac{12}{5}, -1\rangle$, a multiple of $\langle 6, 12, -5\rangle$.

22 MCQ

Apply Gram–Schmidt to $\mathbf{x}_1 = \langle -2, 3, 0\rangle$, $\mathbf{x}_2 = \langle 1, 3, 2\rangle$. Which vector is parallel to the resulting $\mathbf{v}_2$?

- **A** $\langle 1, 3, 2\rangle$
- **B** $\langle -\frac{1}{13}, \frac{60}{13}, 2\rangle$
- **C** $\langle 27, 18, 26\rangle$ ✓
- **D** $\langle -2, 3, 0\rangle$

WORKED SOLUTION & VERIFICATION

$\mathbf{v}_2 = \mathbf{x}_2 - \dfrac{\mathbf{x}_2 \cdot \mathbf{x}_1}{\mathbf{x}_1 \cdot \mathbf{x}_1}\mathbf{x}_1 = \langle \frac{27}{13}, \frac{18}{13}, 2\rangle$, a multiple of $\langle 27, 18, 26\rangle$.

23 MCQ

Apply Gram–Schmidt to $\mathbf{x}_1 = \langle -2, 2\rangle$, $\mathbf{x}_2 = \langle -3, 2\rangle$. Which vector is parallel to the resulting $\mathbf{v}_2$?

- **A** $\langle -3, 2\rangle$
- **B** $\langle -\frac{11}{2}, \frac{9}{2}\rangle$
- **C** $\langle -1, -1\rangle$ ✓
- **D** $\langle -2, 2\rangle$

WORKED SOLUTION & VERIFICATION

$\mathbf{v}_2 = \mathbf{x}_2 - \dfrac{\mathbf{x}_2 \cdot \mathbf{x}_1}{\mathbf{x}_1 \cdot \mathbf{x}_1}\mathbf{x}_1 = \langle -\frac{1}{2}, -\frac{1}{2}\rangle$, a multiple of $\langle -1, -1\rangle$.

24 MCQ

Apply Gram–Schmidt to $\mathbf{x}_1 = \langle 3, 3\rangle$, $\mathbf{x}_2 = \langle -2, -1\rangle$. Which vector is parallel to the resulting $\mathbf{v}_2$?

- **A** $\langle -1, 1\rangle$ ✓
- **B** $\langle 3, 3\rangle$
- **C** $\langle -2, -1\rangle$
- **D** $\langle -\frac{7}{2}, -\frac{5}{2}\rangle$

WORKED SOLUTION & VERIFICATION

$\mathbf{v}_2 = \mathbf{x}_2 - \dfrac{\mathbf{x}_2 \cdot \mathbf{x}_1}{\mathbf{x}_1 \cdot \mathbf{x}_1}\mathbf{x}_1 = \langle -\frac{1}{2}, \frac{1}{2}\rangle$, a multiple of $\langle -1, 1\rangle$.

25 MCQ

Apply Gram–Schmidt to $\mathbf{x}_1 = \langle 3, 3\rangle$, $\mathbf{x}_2 = \langle -3, 1\rangle$. Which vector is parallel to the resulting $\mathbf{v}_2$?

- **A** $\langle 3, 3\rangle$
- **B** $\langle -3, 1\rangle$
- **C** $\langle -1, 1\rangle$ ✓
- **D** $\langle -4, 0\rangle$

WORKED SOLUTION & VERIFICATION

$\mathbf{v}_2 = \mathbf{x}_2 - \dfrac{\mathbf{x}_2 \cdot \mathbf{x}_1}{\mathbf{x}_1 \cdot \mathbf{x}_1}\mathbf{x}_1 = \langle -2, 2\rangle$, a multiple of $\langle -1, 1\rangle$.